

Reflective Journal – NewsBot Midterm

Group 4 – ITAI 2373: Natural Language Processing

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Working on this midterm project gave our team a chance to see how everything we've learned in class actually comes together in a real workflow. We've used NLP tools before in smaller assignments, but this was the first time we had to walk through an entire pipeline from raw data all the way to insights. It pushed us to think not just about running code, but about building something that makes sense, stays organized, and tells a clear story. We also had to step into a quality-control mindset because the early setup from Role 1 wasn't structured well, and Role 2 had to redo several steps to stabilize the workflow. That process required us to understand the entire project deeply enough to fix issues, reorganize the notebook, and make sure everything ran smoothly. In the end, that collaboration became one of the most valuable parts of the experience — learning how to clean up and align different contributions while still delivering a complete, professional result.

One of the biggest lessons we learned was how important preprocessing is. When we looked at the raw dataset, it was messy, repetitive, and full of noise. Once the text was cleaned, lemmatized, and vectorized, the patterns started to make sense. We could see which words mattered, which categories were dominant, and how the writing style changed across topics. The TF-IDF analysis helped highlight the terms that carried meaning, and the POS and dependency parsing showed how news articles are structured. We didn't expect to enjoy the linguistic side of the project as much as we did, but it helped us understand why NLP models behave the way they do. It also made the later steps — like classification and sentiment analysis — feel more grounded because we knew exactly what data the model was learning from.

The classification model was a learning experience for the team. The accuracy was decent, but the imbalance between categories made it clear why some labels performed better than others. Politics and Health had enough examples for the model to learn from, while categories like Tech and Sports had fewer samples. Seeing precision and recall drop to zero for those classes reminded us that machine learning depends heavily on data quality and representation. If we had more time, we would experiment with class balancing techniques or try more advanced models like SVM or transformer-based approaches. Even so, the model still gave us useful insights about which categories were easiest to classify and which ones needed improvement. It also taught us

how to interpret evaluation metrics beyond accuracy — understanding precision, recall, and F1-score helped us see the trade-offs between false positives and false negatives.

The sentiment analysis and named entity recognition were two of the most engaging parts of the project because they made the data feel more human. Sentiment analysis showed that most news writing is neutral, which aligns with journalistic standards. NER helped highlight the people, places, and organizations that appear most often in the dataset. Seeing certain names and topics repeated across categories helped us understand how NLP can support real-world applications like media monitoring and trend tracking. These steps connected the technical side of the project to practical insights. We also realized how combining linguistic features with machine learning can create a more complete picture of how information flows through media.

As a team, we tried to communicate effectively and divide tasks based on strengths. Some of us focused on coding and debugging, while others handled documentation, analysis, and interpretation. We had to coordinate different roles and make sure our work aligned with the overall project goals. There were moments of frustration — especially when earlier code didn't run or produce unexpected results — but those challenges allowed Igwe to grow. Although it was a lot on her, she learned how to troubleshoot effectively, how to respect each other's contributions, and how to maintain consistency in a shared notebook. Collaboration became just as important as technical skills.

Overall, this project taught us how to think like a team building a real NLP system, not just completing an assignment. We had to troubleshoot, reorganize, and make decisions that affected the entire workflow. Late in the project, we also learned how important it is to document work clearly so that others can follow it. If we were to continue improving this project, we would focus on strengthening the classification model, balancing the dataset, and exploring more advanced techniques. But even as it stands, we're proud of how the project came together. It reflects not just what we learned in class, but how we collaborated, adapted, and took ownership of a complex task together.